

Alwyn Sajan

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GitHub: <https://github.com/alwynsajan> | Portfolio: <https://alwynsajan.github.io/Resume/>

Skills

- **Programming Languages:** Python, C, C++, JavaScript
- **Web Development:** Django, Flask, React, HTML, CSS
- **Databases:** SQL, MongoDB, SQLAlchemy (ORM)
- **DevOps:** Docker, Git, CI/CD (basic)
- **Cloud:** AWS (EC2, S3, Lambda)
- **AI / ML:** Machine Learning, Deep Learning, Natural Language Processing, RAG Pipelines, Large Language Models, Multi-Agent Systems

Work Experience

Department of Health (WA Health), Australia

Nov 2025 – Dec 2025

Data Systems List Power Apps Intern, On-site

- Developed a Power Apps solution integrated with SharePoint to streamline management of systems, documents, changes, issues, and tasks.
- Implemented dynamic data filtering, program-based navigation, and status-driven color coding to enhance workflow efficiency.
- Enabled creation and editing of changes, issues, and tasks with detailed record views and status-based prioritization.
- Improved usability, accuracy, and operational efficiency, enhancing system oversight and team collaboration.

AlgoVision Technologies Pvt Ltd, India

Jul 2022 – Jun 2024

Software Engineer

- Enhanced and maintained large Python and C++ codebases, delivering stable features and improving overall system performance and reliability.
- Developed and optimized backend services using Python, Flask, SQL, and MongoDB, improving API responsiveness and data processing efficiency.
- Built internal automation and workflow tools using Tkinter, reducing manual effort and improving operational productivity for engineering teams.
- Refactored and modernized legacy C++ and Python modules by integrating new features, improving runtime efficiency, and ensuring adherence to coding standards.
- Designed, implemented, and deployed REST APIs enabling secure, scalable, and consistent communication between frontend and backend systems.

Projects

- **UWA Logistics Agent - Graph-Orchestrated Multi-Agent AI Platform (UWA AI Club)**
Developed a stateful, multi-agent AI platform to automate room bookings, event management, poster generation, notifications, and interest-based recommendations. Leveraged LangGraph with conditional routing and persistent memory to orchestrate specialized ReAct agents, enabling dynamic intent classification and seam-

less multi-session interactions. Built a modern frontend using React and Vite, and integrated a Dockerized full-stack deployment with MariaDB backend and Google Calendar synchronization for reliable, scalable, and production-ready workflows.

Key Technologies: Python, Django, React, Vite, LangGraph, LangChain, OpenRouter LLM, MariaDB, Docker, REST APIs, Google Calendar API

- **Customer Relationship Management (CRM) System - Freelance Project, Australia**

Designed and developed a modern, end-to-end CRM desktop application using Python and PySide6 to streamline core business operations. Implemented modules for invoice generation and printing, sales recording, sales history tracking, dynamic sales report generation, product stock management, barcode scanning for cart management, and reprinting of previous sales. Leveraged OOP principles and relational database management to ensure scalable architecture, robust data integrity, and maintainable codebase.

Key Technologies: Python, PySide6, MariaDB, RDBMS, OOP

- **WalletWhiz - Personal Finance Web Application (UWA, Australia)**

Created a full-stack personal finance platform using Flask that enables users to manage budgets, visualize expenses, and download financial summaries. Built interactive dashboards and automated spending insights using AJAX, jQuery, and SQLAlchemy. Ensured application reliability with extensive unit tests and Selenium-based end-to-end test coverage.

Key Technologies: Flask, Python, SQLAlchemy, HTML, CSS, JavaScript, jQuery, AJAX, Jinja, Unit Testing, Selenium Testing

- **Multiple Camera Tracking - AllgoVision Technologies**

Enhanced a production-grade multi-camera tracking system used for detecting and following vehicles and people across multiple feeds. Improved legacy C++ and Python modules by adding new features, optimizing compute performance, and reducing processing latency. Supported SQL-based data handling and contributed to modularizing and containerizing components using Docker for more reliable deployment workflows.

Key Technologies: Python, C++, SQL, OOP, Docker

Education

- **Master of Information Technology** 2024 – 2026
University of Western Australia, Australia
GPA: 6.083/7
- **B.Tech in Electronics and Communication** 2018 – 2022
Thangal Kunju Musaliar College of Engineering, India
CGPA: 9.02/10

Certificates

- **Machine Learning Specialization** - Coursera
- **Joy of Computing Using Python** - NPTEL (90%)
- **Python Basics** - HackerRank
- **Responsive Website Basics (HTML, CSS, JS)** - Coursera